David Reyes Alés

Backend Software Engineer · Distributed Systems · Real-time

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Profile

Backend Software Engineer with 2.5+ years building real-time distributed systems and microservices in production. Backend experience across the JVM and Node.js ecosystems — Java, Spring Boot and NestJS — alongside containerisation, CI/CD and SQL/NoSQL data stores. Track record of taking ownership beyond assigned scope, from architecture to company-wide infrastructure. Also hands-on with applied ML/NLP (paper published at IberLEF 2026).

Work Experience

Software Developer — March 2026 - July 2026
The Reuse Company · UC3M — Madrid, Spain

Built a **real-time collaborative SysML v2 modelling tool** (industry research project, MSc thesis) in an Agile team: ASP.NET + PostgreSQL backend and HocusPocus (Yjs CRDTs) for **conflict-free multi-user editing**.

`React/NextJS` `ASP.NET` `PostgreSQL` `HocusPocus` `SysMLv2` `Agile`

Full-Stack Developer — 2023 – 2025, (last months alongside MSc studies)
Axion · Telecommunications — Spain

Joined a 3-person team to rebuild from scratch a nationwide real-time monitoring platform for broadcast infrastructure — a project that had failed over 1.5 years of prior attempts. Delivered in 6 months; adopted by 100+ operations users, after all previous versions had been rejected.

- **Performance:** Identified CouchDB (NoSQL) bottleneck, introduced **Redis Stack** as caching layer (validated via POC), reducing load times **50–60%** (10–11s → 4–5s). Also enabled HTTP compression at zero infrastructure cost.
- **CI/CD:** Built GitLab CI/CD pipeline from scratch on own initiative, cutting deployment time from **5–8h to 40–60 min** (85%+ reduction), with automated test suites catching real production bugs before release.
- **Auth microservice:** Expanded an assigned login module into a centralised authentication microservice for the entire organisation, serving **100+ users** across multiple applications — a decoupled, centralised service design.
- **Map component:** Designed an Angular + Leaflet map handling thousands of real-time assets.
- **Mentoring:** Onboarded an intern; built backend/frontend templates with built-in security and CI/CD.

`NestJS` `Node.js` `TypeScript` `Java` `Spring Boot` `PostgreSQL` `CouchDB` `Redis Stack` `Docker` `GitLab CI/CD` `Angular` `Microservices` `REST APIs` `Linux`

Education

MSc in Computer Science and Technology — 2025 – 2026
Universidad Carlos III de Madrid — Madrid, Spain

BSc in Software Engineering — 2020 – 2025
Universidad de Sevilla — Seville, Spain

Publications

Multi-Label Emotion Detection in Early Modern Spanish Correspondence. **D. Reyes Alés**, A. I. Arteaga, S. Peñafiel Calderón, A. Verdugo Valerio. HISEMOTIONS @ IberLEF 2026, CEUR-WS Vol. 4273. [PDF]

Technical Skills & Languages

Backend	Java · Spring Boot · NestJS · Node.js · REST APIs · Microservices
Data & Infra	PostgreSQL · MySQL · Redis Stack · CouchDB · Docker · GitLab CI/CD · Linux
ML / NLP	Python · PyTorch · Hugging Face Transformers · scikit-learn
Other	Angular · React/Next.js · JavaScript/TypeScript · Automated testing · Agile
Languages	Spanish (native) · English B2 First (FCE), Cambridge, 2023